Plane-Aware Structure-from-Motion

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Abstract

This paper introduces Plane-Aware Structure-from-Motion (PASfM), which leverages dominant planar structures prevalent in real-world scenes as explicit structural priors to improve the planarity (plane adherence) of reconstructed 3D geometry while preserving camera pose quality. Starting from a COLMAP reconstruction, PASfM estimates plane candidates via homographies, globally merges planes that are consistently observed across multiple views, and refines them through outlier removal. Plane-assigned points are then refined using Dynamic Snap, which progressively pulls points toward their associated planes while guarding against divergence. Experiments on ETH3D show AUC performance comparable to COLMAP, along with tighter alignment to dominant planar structures. Across nine scenes, PASfM reduces both the mean and variance of point-to-plane distances in five scenes and reduces the mean in two additional scenes, while the remaining scenes exhibit degraded plane-distance statistics, indicating scene-dependent but measurable gains in plane adherence without compromising pose quality.

1 Introduction

Structure-from-Motion (SfM) is a technique that simultaneously estimates camera poses and reconstructs 3D scene structure from multiple 2D images, and it is widely used in areas such as robot vision and AR/VR. Traditional pipelines represented by COLMAP[1] have provided high reliability through robust processes that combine feature-based matching, geometric verification, and global optimization. As a result, even years after the introduction of COLMAP and GLOMAP[2], COLMAP remains a dominant baseline for real-world scene reconstruction. While recent deep learning–based methods such as MASt3R[3], VGGT[4], and π^3[5] have shown strong performance and can produce dense point clouds, they have not yet fully displaced traditional pipelines, indicating that important practical challenges still remain.

However, conventional approaches such as COLMAP still exhibit fundamental limitations. In real-world scenes, feature matching quality is often degraded by factors such as repetitive patterns, textureless regions, reflective or transparent materials, occlusions, and illumination changes. As a result, reconstruction performance becomes highly scene-dependent and can vary substantially across different environments. Since classical SfM operates on sparse point correspondences and optimizes them independently, it provides no direct mechanism to impose structural regularities in the scene as constraints.

In contrast, humans interpret scenes not primarily through isolated point measurements, but through higher-level geometric regularities. For instance, floors and walls in indoor environments, as well as building facades and window frames in outdoor scenes, frequently exhibit strong planar structure. Such planar regularities can provide robust constraints and help stabilize reconstruction, particularly in regions where texture cues are weak or ambiguous.

Building on these observations, I propose a Plane-Aware Structure-from-Motion (PASfM) pipeline. The goal of PASfM is to improve the accuracy and stability of recovered camera poses and 3D points by explicitly leveraging planar structures prevalent in real-world scenes. To this end, PASfM encourages points on planar surfaces to adhere closely to their associated planes. PASfM takes a COLMAP-based reconstruction as its initial estimate and extends it by detecting, refining, and integrating scene planes into the optimization process.

Specifically, the proposed approach (i) estimates planar candidates within individual images using homographies, (ii) globally merges planes that are repeatedly observed across multiple images to construct consistent plane models, and (iii) refines these planes by removing outliers based on the initial reconstruction. Subsequently, (iv) bundle adjustment is performed and applying plane-guided updates to the associated 3D points. In addition, PASfM introduces auxiliary stabilization strategies. Dynamic Snap progressively moves 3D points toward associated planes to mitigate optimization divergence or distortion that can arise from overly strong planar constraints, thereby ensuring robust convergence.

2 Problem Definition

In traditional Structure-from-Motion, passing geometric verification and being labeled as an inlier does not necessarily imply that the corresponding feature provides a strong constraint on the camera pose during bundle adjustment. This gap arises because reprojection error measures only 2D consistency on the image plane, not how well the underlying parameters are determined. In particular, a small reprojection residual does not guarantee that the estimated camera poses and 3D points are unique, stable, or tightly constrained. While modern pipelines employ rigorous geometric verification to suppress unreliable matches, this robustness often comes with an inherent trade-off: removing ambiguous correspondences can leave too few well-distributed matches, increasing the risk of image registration failure. In this section, I analyze the technical causes of this phenomenon, where geometrically verified correspondences still behave as weak constraints and induce instability.

2.1 Triangulation Uncertainty from Limited Parallax

The most direct geometric cause of weak constraints is depth uncertainty caused by insufficient parallax during triangulation. When camera baselines are small, the scene is distant, or the motion is predominantly forward, the intersection angle between viewing rays becomes very small. In this case, many 3D points can represent nearly the same 2D measurements, and the feasible region expands strongly along the depth direction. As a result, a correspondence may remain an inlier with low reprojection error, yet the residual changes only weakly for large depth changes. This yields small residuals but high 3D uncertainty, and it reduces how effectively BA can lock the pose using those observations.

2.2 Failure Modes under Unstable and Spatially Biased Matching

In scenes dominated by large planar surfaces or adverse imaging conditions—such as reflections, transparency, and severe illumination changes—feature correspondences become unreliable. Descriptor distinctiveness decreases and photometric assumptions break down, so many tentative matches are rejected during geometric verification. The surviving correspondences are often sparse and concentrated in a few textured regions.

Under this lack of constraint coverage, the reconstruction becomes disproportionately driven by a limited set of constraints. In such cases, even a few incorrect correspondences can dominate the optimization because there are insufficient independent, well-distributed matches to suppress their influence. As a result, the reconstruction may drift or converge to an erroneous configuration, and image registration can fail despite the presence of many candidate matches that appear locally consistent.

2.3 Limitations of Reprojection Error

Traditional SfM is driven by minimizing reprojection error on the image plane. This objective enforces 2D consistency, but it does not directly penalize physically implausible distortions in 3D. When viewpoints are sufficiently diverse and provide strong parallax, the multiview geometry becomes well constrained, and many ambiguities are naturally resolved. However, the reconstruction can remain highly ambiguous when the scene exhibits degenerate structure, such as being dominated by planar surfaces or repetitive patterns. The same issue also arises when there are many images but limited viewpoint diversity, as discussed in Section 2.1. Consequently, drift or distorted 3D configurations may be accepted as plausible solutions because the objective provides only weak resistance along poorly constrained directions.

3 Motivation

The goal of this project is to enhance reconstruction stability and structural consistency after successful image registration by ensuring that reconstructed 3D points adhere to the scene's geometric structure, especially planarity on large surfaces.

COLMAP and GLOMAP demonstrate exceptional robustness in environments that guarantee sufficient overlap and high-quality feature points. However, in real-world settings, factors such as repetitive textures, textureless planar structures, reflective or transparent materials, occlusions, and illumination changes frequently cause the reliability of

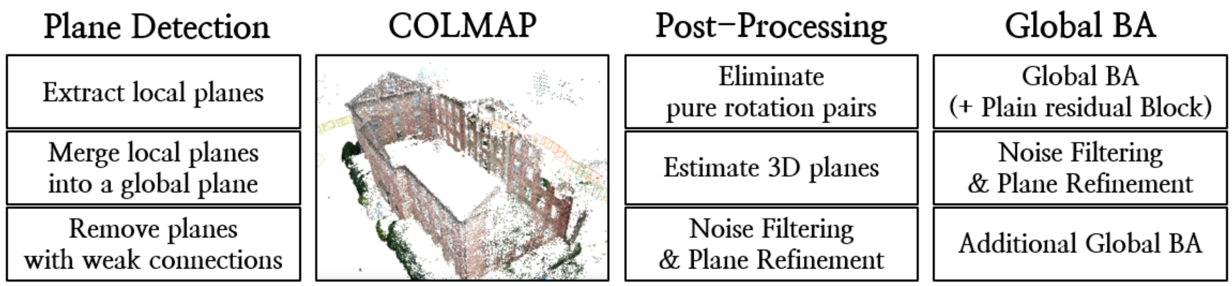

Figure 1: The overall pipeline of PASfM.

feature matching to become spatially heterogeneous. In such environments, even inliers that pass geometric verification often remain as weak constraints that lack sufficient binding force during the optimization process. Consequently, even if camera pose estimation converges, problems such as accumulated local drift, structural distortion, and instability in scale and shape arise, leading to significant deviations in reconstruction quality depending on scene characteristics. This limitation stems from the fact that traditional SfM inherently relies on minimizing reprojection error based on sparse point correspondences, lacking a mechanism to directly utilize the low-rank structures—specifically, geometric priors like "planes"—that are abundant in man-made environments.

Meanwhile, recently emerged deep learning–based 3D reconstruction methods have expanded feasibility under adverse conditions—such as extremely low overlap or sparse views—where traditional SfM often fails already at the registration stage. However, these approaches still face clear limitations, including constraints on input resolution, prohibitive computational costs, and difficulties in enforcing globally consistent multi-view geometry [6]. Moreover, pose stability tends to degrade as scene scale and complexity increase, making it difficult to match the robustness of classical SfM. Consequently, for high-resolution, large-scale photogrammetry, traditional pipelines remain indispensable due to their rigorous geometric verification, globally consistent alignment, and stable convergence.

Against this backdrop, the Plane-Aware Structure-from-Motion (PASfM) proposed in this study aims to complement existing SfM pipelines rather than replace them. PASfM preserves the robustness of geometric verification and global optimization offered by COLMAP, while incorporating an intermediate-level structural prior—plane geometry—to reinforce reconstruction where traditional SfM becomes vulnerable, such as textureless planar regions, repetitive patterns, and configurations that induce weak observational constraints. As analyzed in Section 2, PASfM mitigates this weak-constraint instability by using dynamic snap to actively pull plane-associated 3D points onto their plane, so that the reconstructed points explicitly form a planar surface.

4 Approach

4.1 Plane Detection

The PASfM pipeline begins by extracting local planes on a per–image-pair basis. A local plane denotes a plane candidate defined only within a specific matched image pair. The key assumption is as follows: if a plane exists in 3D space, then the set of correspondences lying on that plane can be explained by a single homography between the two views. Accordingly, for each image pair, PASfM estimates a homography via RANSAC[7] and takes the model that yields the largest inlier ratio as the local plane candidate. Since multiple planes may coexist in a scene, once a plane is detected, PASfM removes its inlier correspondences and repeats the same procedure on the remaining correspondences.

Threshold selection at this stage is critical. If too many planes are extracted, planes found later tend to be supported by progressively weaker evidence. Conversely, if too few planes are extracted, the total number of feature points participating in dynamic snapping becomes insufficient, weakening the effective binding force. As a result, even if snapping shifts points toward the plane once, subsequent bundle adjustment can pull them back toward their original configurations. In addition, if the RANSAC threshold is set too low, the inlier set becomes overly small and the binding force weakens; if it is set too high, many outliers are admitted, degrading plane quality.

In the next stage, PASfM merges the local planes into global planes at the scene level. A global plane represents a unique plane entity that exists consistently across the entire scene. This leverages the fact that local planes detected from different image pairs may correspond to the same underlying plane. The merging criterion is the number of overlapping features: if two local planes share at least τ_{ov} features, they are connected as candidates of the same global plane, and each connected component is then treated as one global plane. However, naive merging can lead to a contradiction

Algorithm 1 Local Plane Extraction (RANSAC)

Require: Matches $\mathcal{M}$, keypoints $\mathcal{F}$, threshold τ_{px}, min inliers $N_{\min}$, max planes $K_{\max}$
Ensure: FeatureToPlanes
FeatureToPlanes $\leftarrow \emptyset$
for all image pairs (I, J) with sufficient matches **do**
    Extract matched keypoint coordinates p_I, p_J from $\mathcal{F}$ using $\mathcal{M}$
    for $k = 1$ **to** $K_{\max}$ **do**
        $(H, m) \leftarrow$ HomographyRANSAC$(p_I, p_J; \tau_{\text{px}})$
        if $H = \emptyset$ **or** $\sum_i m_i < N_{\min}$ **then**
            break
        end if
        Assign all inlier matches $(m = 1)$ to a new local plane ID $\pi_{I,J,k}$
        Remove inliers from (p_I, p_J) and continue
    end for
end for
return FeatureToPlanes

Algorithm 2 Merge Local Planes into Global Planes

Require: Local assignments FeatureToPlanes, overlap threshold τ_{ov}, min support $\tau_{\min}$
Ensure: FeatureToGlobalPlane
Compute pairwise overlap counts between local planes:
for all features that belong to multiple local planes **do**
    increment overlap for all plane pairs
end for
Build an undirected graph G over local planes:
for all plane pairs (p_i, p_j) **do**
    if Overlap$(p_i, p_j) \geq \tau_{\text{ov}}$ **then**
        add edge (p_i, p_j)
    end if
end for
Get connected components of G and assign each component a global plane ID g
FeatureToGlobalPlane $\leftarrow \emptyset$
Support$[g] \leftarrow 0$ ▷ support counter per global plane
for all features (id, u) with local plane list $\mathcal{P}$ **do**
    $\mathcal{G} \leftarrow \{$ ConnectedComponentID$(p) \mid p \in \mathcal{P} \wedge p \in G \}$
    if $|\mathcal{G}| = 1$ **then**
        $g \leftarrow$ the only element in $\mathcal{G}$
        FeatureToGlobalPlane$[id][u] \leftarrow g$
        Support$[g] \leftarrow$ Support$[g] + 1$
    end if
end for
Keep only global planes with Support$[g] \geq \tau_{\min}$ and discard the rest
return filtered FeatureToGlobalPlane

where a single 3D point is assigned to multiple planes simultaneously. To prevent this, PASfM enforces that each 3D point belongs to exactly one global plane, and removes points referenced by two or more planes as outliers. Finally, to retain only planes with sufficient support after filtering, PASfM passes to the next stage only those global planes whose feature count exceeds τ_{min}. Importantly, plane detection in PASfM is not intended to exhaustively recover every plane in the scene; rather, it serves as a stability-oriented safeguard that minimizes false positives and retains only trustworthy planes to prevent breakdowns in subsequent optimization.

4.2 Post-Processing

The next step removes image pairs that are close to pure rotation. Pure rotation is the most dangerous counterexample for homography-based plane detection. Because even in the absence of any actual plane, nearly all correspondences can still be explained by a single homography. This case is difficult to identify reliably using only the pixel coordinates of the two images. Therefore, PASfM leverages the approximate camera poses and 3D points provided by COLMAP to

Algorithm 3 Refine 3D Planes (RANSAC + SVD)

Require: Reconstruction $\mathcal{R}$, assignments FEATURETOGLOBALPLANE, inlier threshold τ, confidence γ, min support $T_{\min}$, max iters K
Ensure: PLANEMODEL, PLANETOPIDS
(PLANETOPOINTS, PLANETOPIDS) $\leftarrow$ COLLECTPLANEPOINTS($\mathcal{R}$, FEATURETOPLANE)
for all global planes g in PLANETOPOINTS **do**
    $\mathbf{X} \leftarrow$ PLANETOPOINTS$[g]$
    $(\pi_{\text{ransac}}, \mathcal{I}) \leftarrow$ FITPLANERANSAC$(\mathbf{X}; \tau, \gamma, K)$
    INLIERPIDS $\leftarrow \{$ PLANETOPIDS$[g][i] \mid i \in \mathcal{I} \}$
    if $|$INLIERPIDS$| \geq T_{\min}$ **then**
        $\pi \leftarrow$ FITPLANESVD$(\mathbf{X}[\mathcal{I}])$ ▷ SVD refit on inliers
        PLANEMODEL$[g] \leftarrow \pi$
        PLANETOPIDS$[g] \leftarrow$ INLIERPIDS
    end if
end for
return (PLANEMODEL, PLANETOPIDS)

compute the distribution of triangulation angles for points observed in each image pair. Key criterion is whether the 25th percentile of triangulation angles falls below 1°. I use the 25th percentile to prevent a small number of correspondences with a good baseline from masking a near-pure-rotation pair.

After filtering near-pure-rotation pairs and refining the set of global plane candidates, plane models are estimated in 3D space. This stage uses the 3D points from the COLMAP reconstruction and their global-plane assignments as input. A coarse plane is first obtained via RANSAC: three points are sampled to compute the plane normal and offset, and the best model is selected by maximizing the inlier count under a point-to-plane orthogonal distance threshold. The plane is then refit by SVD using only the inlier points. This reduces RANSAC sampling noise and yields a more accurate plane normal for downstream steps. This reliability is critical because later stages move 3D points explicitly along the estimated normal direction, so errors in the estimated normal directly cause incorrect point updates along that direction.

4.3 Global BA

A straightforward approach is to add the BA objective with a point-to-plane orthogonal distance term. For the set of planar points $\mathcal{P}$, the following term can be included:

$$E_{\text{plane}} = \sum_{i \in \mathcal{P}} r_i^2 = \sum_{i \in \mathcal{P}} \lambda(\mathbf{n}^\top \mathbf{X}_i + d)^2. \tag{1}$$

Since this penalty can be reduced most easily by shrinking the overall scale (i.e., pulling cameras and points closer together), two cameras must be fixed.

Despite its simplicity, this formulation has a critical drawback: optimization becomes slower. Plane constraints couple points through shared plane parameters, which weakens the benefits of the Schur complement and reduces the efficiency of standard BA.

PASfM therefore avoids injecting a hard plane residual into BA. Instead, it interleaves standard BA with an external point update that moves plane-assigned points toward their planes. The update is defined by the following snapping rule:

$$\mathbf{X}' = \mathbf{X} - \alpha(\mathbf{n}^\top \mathbf{X} + d)\mathbf{n}, \tag{2}$$

where α controls the snapping strength. While $\alpha = 1$ projects points exactly onto the plane, this can increase reprojection error under real, noisy measurements. PASfM thus starts from a small α and applies the update gradually.

Instead of fixing α heuristically in advance, PASfM employs a dynamic snapping strategy that adjusts α based on the change in reprojection error. After applying snapping with the current α_{i-1}, α is reduced if the reprojection error increases, and slightly increased if the error decreases, following a line-search-style procedure. This design has two benefits. First, because plane constraints are not embedded in BA, the Schur-complement structure is preserved and optimization remains fast. Second, reprojection error acts as an acceptance signal: when snapping increases the error, the update is automatically damped by reducing α. In other words, PASfM does not aim for forced projection that exactly satisfies planarity; it aims for a safe update that shifts points toward planar structure without violating observational consistency.

Algorithm 4 Dynamic Snap

Require: baseline error E_{base}, init step α_0, tolerance δ, shrink β, bounds $[\alpha_{\min}, \alpha_{\max}]$, max trials N
Ensure: $(\textsc{Accepted}, \alpha^\star)$

$\alpha \leftarrow \text{clip}(\alpha_0, 0, \alpha_{\max})$
$\textsc{Accepted} \leftarrow$ **false**, $\alpha^\star \leftarrow 0,\ g \leftarrow 1/\beta$
for $t = 1$ **to** N **do**
    if $\alpha \leq \alpha_{\min}$ **then**
        break
    end if
    apply fixed snap with α
    compute E_{new}
    if $E_{\text{new}} \leq (1 + \delta)\, E_{\text{base}}$ **then**
        $\textsc{Accepted} \leftarrow$ **true**, $\alpha^\star \leftarrow \alpha$
        $\alpha \leftarrow \min(\alpha \cdot g, \alpha_{\max})$
    else
        if $\textsc{Accepted}$ **then**
            break
        end if
        $\alpha \leftarrow \alpha \cdot \beta$
    end if
end for
return $(\textsc{Accepted}, \alpha^\star)$

Finally, the overall flow repeats n times: run a small number of BA iterations, refine plane models, apply snapping to move points toward planes, and run BA again. This iterative process avoids destabilizing the optimizer with a single large update and forms a feedback loop in which updated points and poses enable more accurate plane estimation. Moreover, when combined with preceding noise filtering, the set of points being snapped becomes progressively cleaner, reducing the risk of converging points toward an incorrect plane. The core of PASfM is thus to repeatedly apply plane-based external point updates, guarded by reprojection consistency, to encourage structural alignment of planar points, while retaining the fast and stable optimization backbone of traditional SfM and mitigating structural collapse in low-texture regions.

4.4 Weighted BA

In addition to plane information, an attempt was made to incorporate feature-level reliability. SuperGlue provides a matching score for each correspondence, which can be interpreted as a confidence measure. Under this assumption, a single weight is assigned to each 3D point by taking the maximum score over its track, and this weight is then used to scale the reprojection-error term in bundle adjustment.

5 Results

5.1 Experimental Environment

- **Platform:** Vessl AI
- **GPU:** NVIDIA GeForce GTX 1080
- **Dataset:** ETH3D MVS (DSLR)[8]
- **Feature extraction:** SuperPoint
- **Feature matching:** SuperGlue
- **Matching pairs:** All image pairs
- **Evaluation metrics:** (i) camera pose AUC at $1°/3°/5°$, (ii) distance distribution to the top-5 planes using the ETH3D scan data as the reference, (iii) plane offset error and normal cosine similarity with respect to the ground truth.

	AUC @ 1°				AUC @ 3°				AUC @ 5°			
Scene	COLMAP	COLMAP+WBA	PASfM	PASfM+WBA	COLMAP	COLMAP+WBA	PASfM	PASfM+WBA	COLMAP	COLMAP+WBA	PASfM	PASfM+WBA
boulders	74.2	69.7	74.5	71.3	91.2	89.5	91.2	90.2	94.7	93.7	94.7	94.6
bridge	75.5	17.3	75.4	59.7	91.6	23.9	91.5	77.2	94.9	27.8	94.9	81.6
courtyard	36.2	17.1	36.3	24.1	56.2	31.4	56.3	46.7	61.9	38.1	61.9	53.4
delivery area	50.4	50.3	50.6	50.6	70.7	70.1	70.7	70.7	75.3	75.1	75.3	75.3
door	85.4	85.1	85.1	50.6	94.1	94.1	91.8	91.8	96.5	96.4	96.2	96.2
electro	57.1	50.3	58.1	57.1	82.5	50.8	82.8	80.1	88.9	56.2	89.1	85.9
facade	54.9	4.2	54.6	51.9	66.6	9.1	66.5	64.1	69.8	14.6	69.7	67.0
kicker	61.1	0.4	61.1	55.1	85.4	1.8	85.5	80.0	91.0	4.9	91.1	86.2
living room	73.5	0.1	73.2	23.3	90.4	0.1	90.3	31.0	94.1	0.1	94.1	34.8
observatory	38.3	0.2	38.7	38.7	73.0	0.3	73.2	73.2	83.1	0.3	83.3	83.2
relief	61.4	0.2	60.2	46.8	85.8	0.8	85.5	80.1	91.3	1.2	91.2	87.8
relief 2	56.8	0.2	56.5	17.5	83.8	0.3	83.5	26.1	90.2	0.4	89.9	28.9
terrace	53.9	53.9	55.1	54.3	81.6	81.6	82.5	81.6	88.5	88.5	89.2	89.0
terrace 2	87.8	87.5	87.7	87.5	96.0	94.3	95.9	95.1	97.6	95.5	97.6	95.6
terrains	65.0	1.5	64.8	21.5	87.5	3.6	87.4	33.4	92.4	5.0	92.3	37.7
Average	62.1	29.2	62.1	47.3	82.4	36.8	82.3	68.1	87.4	39.9	87.4	73.2

Table 1: Camera pose accuracy (AUC) at $1^\circ/3^\circ/5^\circ$ on ETH3D MVS(DSLR). The best score in each column is highlighted in green.

5.2 Camera Pose Accuracy

Camera pose accuracy was evaluated on ETH3D MVS (DSLR) using AUC at $1^\circ/3^\circ/5^\circ$, as reported in Table 1. Scenes not listed in Table 1 were excluded because no global plane could be extracted for those scenes, making comparison infeasible. On average, PASfM achieves nearly the same pose accuracy as COLMAP. At 1°, both methods obtain a mean AUC of 62.1; at 3°, COLMAP reaches 82.4 while PASfM attains 82.3; and at 5°, both again achieve 87.4. These results indicate that incorporating plane-guided updates does not meaningfully degrade overall pose alignment performance.

In contrast, configurations that combine weighted BA (WBA) based on matching scores exhibit a substantial decrease in pose AUC. The mean AUC of COLMAP+WBA drops to 29.2/36.8/39.9 at $1^\circ/3^\circ/5^\circ$, and PASfM+WBA partially recovers to 47.3/68.1/73.2, but still falls short of the average performance achieved by COLMAP or PASfM alone. This suggests that the matching-score weighting is more likely to harm the numerical stability of BA than to provide effective constraints for pose optimization.

5.3 Distance distribution to the top-5 plane

To find GT planes from the ETH3D laser scans, the scan point clouds were aligned and merged into a global frame, and dominant planes were extracted via RANSAC to generate per-point plane labels. The labeled scan points were then projected into each image using the ETH3D ground-truth poses and intrinsics to obtain pixel-wise GT plane label maps. For both COLMAP and PASfM, each reconstructed 3D point was assigned to a GT plane by majority voting over the pixel labels along its track, and assignments were accepted only when the observation count and vote-ratio thresholds were met. For a fair comparison, only points assignable under identical criteria in both reconstructions were retained, and evaluation was restricted to points assigned to the same GT plane. After each reconstruction was aligned to the GT camera centers with a Sim(3) transform, planarity was quantified by comparing point-to-plane orthogonal distance distributions on the five most supported GT planes.

Table 2 reports the mean and standard deviation of point-to-plane orthogonal distances, quantifying how far plane-assigned 3D points deviate from the GT planes. The table includes only scenes for which ETH3D scan data are available and at least one global plane could be extracted by the pipeline. Overall, PASfM yields a slightly lower mean distance of 1.1201 m than COLMAP, which attains 1.1234 m, indicating a marginal improvement in average plane alignment. At the same time, the standard deviation increases from 1.4309 m to 1.4338 m, showing that the distance distribution becomes slightly more dispersed under PASfM.

This trend varies across scenes. *courtyard*, *relief_2*, *delivery_area*, *electro*, and *kicker* exhibit decreases in both the mean and standard deviation, suggesting consistently improved alignment to the GT planes. In *relief* and *facade*, the mean distance decreases but the standard deviation increases. In contrast, *terrace* and *terrains* show increases in both the mean and standard deviation, indicating that plane-guided updates clearly degrade plane-alignment performance in these scenes.

5.4 Plane Parameter Consistency to Ground Truth

Table 3 evaluates the geometric accuracy of the estimated plane models themselves, rather than point-to-plane adherence. It reports (i) plane offset error to the GT plane (m) and (ii) cosine similarity between the estimated and GT unit

Scene	**COLMAP**		**PASfM**	
	Mean (m)	**Std (m)**	**Mean (m)**	**Std (m)**
courtyard	0.8758	0.5277	0.8599	0.5276
relief	0.0238	0.0997	0.0234	0.1018
relief 2	0.0235	0.0683	0.0227	0.0681
terrace	0.0544	0.0696	0.0586	0.0847
terrains	0.1563	0.4954	0.1580	0.4964
delivery area	0.6892	1.9333	0.6887	1.9332
electro	0.1661	0.3532	0.1654	0.3531
facade	7.1179	7.7802	7.1003	7.7891
kicker	1.0041	1.5505	1.0041	1.5505
Overall	1.1234	1.4309	1.1201	1.4338

Table 2: Orthogonal distance statistics to GT planes in meters. The best score in each column is highlighted in green.

Scene	Plane Offset Error to GT (m)				Normal Cosine Similarity to GT ($\times$100)			
	COLMAP		PASfM		COLMAP		PASfM	
	Mean	Std	Mean	Std	Mean	Std	Mean	Std
courtyard	0.886	0.605	0.873	0.596	99.847	0.079	99.849	0.078
relief	0.101	0.185	0.105	0.196	99.954	0.088	99.955	0.085
relief 2	0.015	0.010	0.011	0.004	99.980	0.024	99.982	0.021
terrace	0.114	0.138	0.140	0.185	99.989	0.014	99.985	0.002
terrains	3.239	6.160	3.243	6.167	76.609	37.782	76.573	37.744
delivery area	2.144	3.532	2.144	3.532	80.012	39.771	80.016	39.761
electro	0.294	0.376	0.293	0.375	99.946	0.067	99.946	0.066
facade	7.282	6.963	7.266	6.955	78.045	37.382	78.006	37.454
kicker	1.710	1.454	1.710	1.454	61.929	35.534	61.937	35.523

Table 3: Scene-wise plane-fit quality against GT planes, measured by plane offset error and normal cosine similarity. The best score in each column is highlighted in green.

normals. Across scenes, normal accuracy is bimodal. In *courtyard*, *relief*, *relief 2*, *terrace*, and *electro*, cosine similarity is saturated near 0.999 for both COLMAP and PASfM, showing that dominant plane orientations are consistently recovered. In *terrains*, *delivery area*, *facade*, and *kicker*, the mean cosine similarity drops to roughly 0.6–0.8 with very large standard deviations (0.35–0.40), indicating unstable plane hypotheses. For these scenes, the limiting factor is therefore plane hypothesis formation and point assignment, not the snapping refinement step.

These results help explain why the orthogonal distance distributions in Table 2 degrade for *terrace* and *terrains*. When the normal cosine similarity is extremely high (around 0.999), the orthogonal distance distribution is largely governed by the plane offset error rather than orientation error. In this case, two surfaces with nearly identical orientation but slightly different depths may be merged into a single plane hypothesis and then snapped.

Consistent with this interpretation, points adhere more closely to the plane in *courtyard*, *relief 2*, and *electro*, where PASfM exhibits smaller offset errors than COLMAP. In particular, although *electro* shows slightly lower normal similarity under PASfM than COLMAP, the similarity is already near 0.999, suggesting that plane offset has a stronger effect on the distance distribution. In contrast, *relief* shows a larger plane offset under PASfM, which likely contributes to the increased standard deviation of orthogonal distances. For *terrace*, both a larger offset error and a lower normal similarity, leading to a larger overall deviation in the orthogonal distance distribution.

When the normal cosine similarity drops to the 0.6–0.8 range, normal misalignment appears to have a stronger influence than offset. In *facade*, COLMAP achieves higher normal similarity than PASfM, but it also exhibits a larger offset error, and PASfM yields a better orthogonal distance distribution overall. Finally, *terrains* shows both larger offset error and lower normal similarity, which is consistent with its larger errors in the orthogonal distance distribution.

Scene	GT Label Consistency (%)
courtyard	99.6
relief	–
relief 2	–
terrace	73.2
terrains	79.8
delivery area	81.5
electro	–
facade	99.8
kicker	–

Table 4: Scene-wise GT label consistency of extracted global planes(%). A dash (–) indicates that none of the extracted global planes in that scene could be matched to any of the top-5 GT planes. In many scenes, the dominant scan-derived GT planes correspond to large walls or floors, whereas PASfM may fail to capture these weakly textured surfaces, making a direct comparison against the top-5 GT planes infeasible.

6 Analysis

6.1 Reflections in windows or mirrors

One failure factor is features induced by reflections in windows or mirrors. In indoor and semi-indoor scenes, reflected objects can still produce coherent correspondences that satisfy homography constraints, so they are mistakenly included in planar hypotheses. As a result, the estimated plane can spill across the window boundary and expand into regions that do not belong to the actual surface, as shown in Fig 3. This issue is not specific to PASfM and also affects feature-based SfM pipelines such as COLMAP. In ETH3D, features observed through windows often correspond to far-field background geometry outside the target scene. In most cases, these should be excluded because they exhibit little parallax and provide weak geometric constraints. Including these features biases plane estimation and degrades the reliability of plane-guided updates, showing that geometric reasoning alone is insufficient unless reflective or transparent regions are handled explicitly.

6.2 Hyperparameter Sensitivity and Limited Robustness

The proposed pipeline relies on many hyperparameters. These include thresholds for homography inlier ratios, minimum plane support, overlap criteria for plane merging, and several parameters that control snap strength and dynamic snapping. Because these parameters interact, the method is stable only within a narrow range of settings. In scenes with strong planar structure, relaxed thresholds often produce several plausible planes. In scenes with weak or ambiguous planarity, slightly stricter thresholds can prevent plane detection entirely. To reduce false positives, I ultimately chose conservative settings. As a result, the pipeline failed to extract any plane in some scenes. Overall, this exposes a clear trade-off between plane coverage and reliability.

6.3 Inability to Resolve Subtle Depth Variations

Homography-based plane detection struggles to separate surfaces that have similar orientations but slightly different depths, as shown in Fig 3. Under noisy correspondences and limited parallax, the homographies induced by these surfaces become nearly identical. As a result, closely spaced but distinct structures are often merged into a single plane. This failure is most common with small-baseline motion, where homographies approach the pure-rotation case. Overall, pairwise homography constraints do not provide enough discrimination to recover fine depth variation.

To support this claim, I measured GT label consistency by re-assigning the 3D points of each extracted global plane to GT planes using rendered GT plane labels. Table 4 reports a scene-wise top-1 ratio, defined as the maximum fraction of points assigned to a single GT plane among the extracted global planes. Notably, *terrace* attains a high normal accuracy in Table 2, yet its point-to-plane orthogonal distance distribution is worse than COLMAP due to an offset mismatch. The low top-1 consistency in Table 4 provides direct evidence for this: many points grouped into a single PASfM plane do not belong to the same physical GT plane, but instead spread across multiple, nearly parallel GT planes. This explains why the estimated normals can remain correct while the plane offsets drift, degrading the orthogonal distance statistics despite seemingly accurate orientations.

6.4 Dependence on Successful Image Registration

Plane-aware optimization operates strictly within the space of already-registered images and reconstructed points. It therefore cannot recover scenes where image registration fails due to low texture, repetitive patterns, or failed geometric verification. Without reliable camera poses, plane extraction, refinement, and snapping cannot influence the reconstruction. In short, plane-aware cues can refine and stabilize an existing solution, but they cannot fix failures in the core SfM stages.

6.5 Failure of Matching-Score–Weighted Bundle Adjustment

The weighted bundle adjustment using SuperGlue matching scores did not produce meaningful gains. Across scenes, the score distribution was heavily concentrated near 1.0, and the scene-wise mean of per-point maximum scores was close to 0.9, as shown in Fig 6. Because the scores have little dynamic range, the weights cannot separate strong correspondences from weak ones. In practice, the objective becomes nearly identical to the unweighted reprojection error, yet the reweighting still alters the optimization enough to worsen convergence. This indicates that raw SuperGlue scores are not well-calibrated confidence values for optimization and should not be used as weights without further normalization or transformation.

To strengthen my analysis, I ran an additional experiment by fixing the weight to $w = 1$ and performing bundle adjustment with the standard L2 reprojection loss. As shown in Table 5, COLMAP+WBA with uniform weights ($w = 1$) produces results that are overall similar to COLMAP+WBA using matching-score weights. This suggests that the SuperGlue matching scores provide insufficiently reliable information as confidence weights for optimization, and that directly using raw matching scores as reprojection weights is unlikely to yield meaningful benefits.

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Appendix

Figure 2: Yellow points represent features belonging to the plane after post-processing, red points indicate outliers identified during the process, and blue points denote features classified as planar by other image pairs.

Figure 3: (Left) Feature points on reflected posters are detected and matched, causing the virtual area behind the window to be extracted as a plane. (Right) Two planes with identical normal vectors but different depths are incorrectly recognized as a single plane.

Figure 4: Curved surfaces or complex geometries are erroneously identified as planar structures.

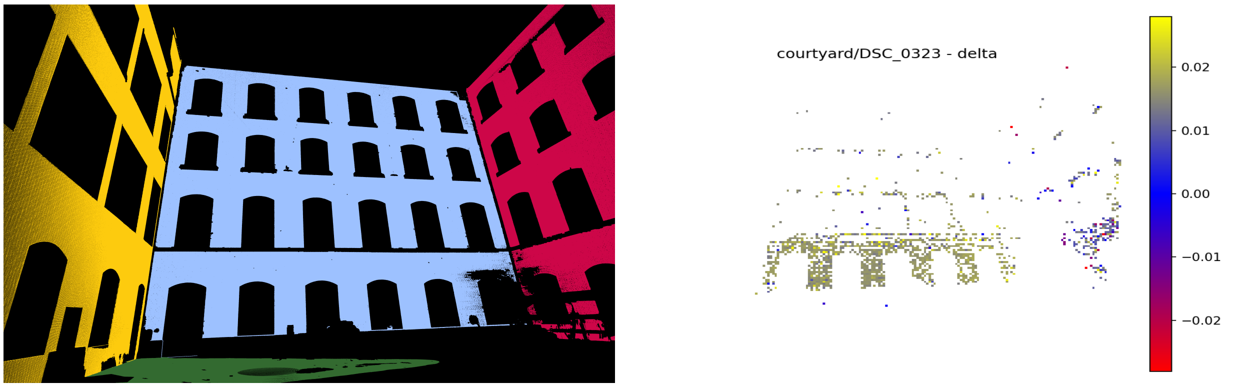
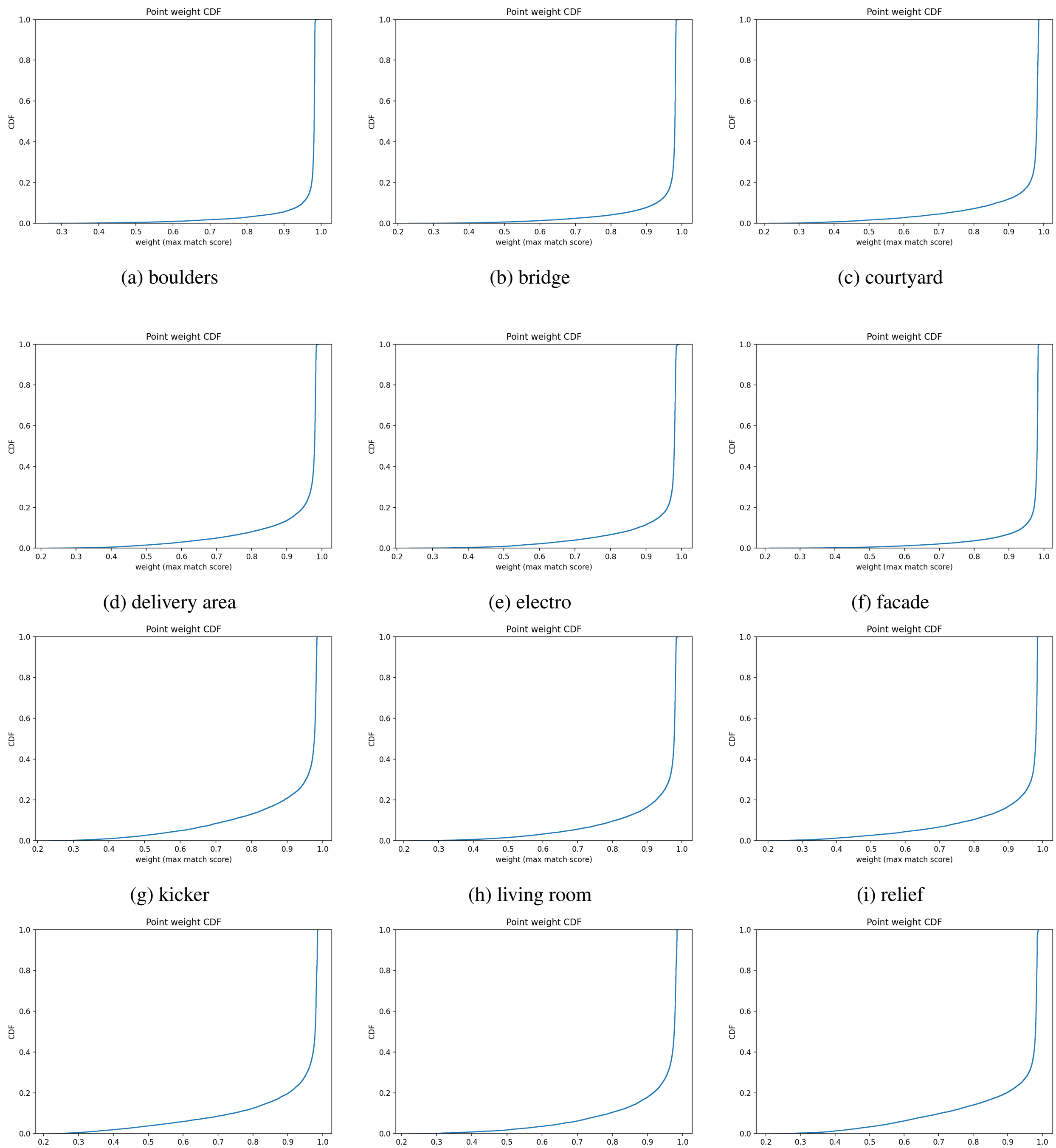


Figure 5: (Left) Visualization of the dominant planes derived from the ETH3D scan data. (Right) Difference in point-to-plane orthogonal distance to the GT planes between COLMAP and PASfM, computed as $\Delta d = d_{\perp}^{\text{COLMAP}} - d_{\perp}^{\text{PASfM}}$
.

	AUC @ 1°			AUC @ 3°			AUC @ 5°		
Scene	COLMAP	COLMAP+WBA	COLMAP+WBA ($w=1$)	COLMAP	COLMAP+WBA	COLMAP+WBA ($w=1$)	COLMAP	COLMAP+WBA	COLMAP+WBA ($w=1$)
boulders	74.2	69.7	69.7	91.2	89.5	89.5	94.7	93.7	93.7
bridge	75.5	17.3	17.3	91.6	23.9	23.8	94.9	27.8	27.8
courtyard	36.2	17.1	17.0	56.2	31.4	31.3	61.9	38.1	38.0
delivery area	50.4	50.3	50.4	70.7	70.1	70.7	75.3	75.1	75.3
door	85.4	85.1	85.4	94.1	94.1	94.1	96.5	96.4	96.5
electro	57.1	50.3	35.2	82.5	50.8	50.8	88.9	56.2	56.1
facade	54.9	4.2	4.1	66.6	9.1	8.6	69.8	14.6	13.9
kicker	61.1	0.4	0.4	85.4	1.8	1.8	91.0	4.9	4.8
living room	73.5	0.1	0.1	90.4	0.1	0.2	94.1	0.1	0.2
observatory	38.3	0.2	0.2	73.0	0.3	0.4	83.1	0.3	0.6
relief	61.4	0.2	0.3	85.8	0.8	0.8	91.3	1.2	1.2
relief 2	56.8	0.2	0.7	83.8	0.3	1.2	90.2	0.4	1.3
terrace	53.9	53.9	53.9	81.6	81.6	81.6	88.5	88.5	88.5
terrace 2	87.8	87.5	87.8	96.0	94.3	96.0	97.6	95.5	97.6
terrains	65.0	1.5	0.9	87.5	3.6	1.9	92.4	5.0	2.7
Average	62.1	29.2	28.3	82.4	36.8	36.8	87.4	39.9	39.9

Table 5: COLMAP is compared against COLMAP+WBA (matching-score weights) and COLMAP+WBA with uniform weights ($w = 1$). The $w = 1$ setting is numerically equivalent to standard unweighted bundle adjustment.



(a) boulders (b) bridge (c) courtyard

(d) delivery area (e) electro (f) facade

(g) kicker (h) living room (i) relief

(j) relief 2 (k) terrace (l) terrians

Figure 6: Cumulative Distribution of Matching Scores per Scene